

Adrián Chalco González

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I am an undergraduate biology student passionate about understanding evolution through comparative genomics, epigenetic regulation and phylogenomics. I seek to expand my knowledge by learning computational, mathematical and statistical tools that allow me to unravel the fundamentals of genome evolution. I am a leader, disciplined, proactive and true to my principles.

EDUCATION

Peruvian University Cayetano Heredia

Jan 2021 – Dec 2025

B.Sc. Biological Sciences

Lima, Peru

- No. 1 Biological Sciences program in Peru
- GPA: 4.0/4.0, 17.3/20.0 in original scale
- Currently **1st** out of 75 students in the annual merit order of the Faculty of Sciences and Engineering.

EXPERIENCE

Research Internship

Jan 2025 – Present

Sedlazeck Lab – Human Genome Sequencing Center, Baylor

Remote

College of Medicine, USA

PI: Fritz Sedlazeck, Ph.D

- Built **Virus2Tree**: a Bash/Python pipeline that turns NCBI coding sequence accessions into an OMA reference database for Read2Tree, bulk downloads SRA runs, and launches end-to-end viral phylogenomics. Automates mature-peptide extraction for precise CDS/peptide datasets, read deduplication, and coverage-aware downsampling.

Research Internship

Jul 2024 – Jan 2025

Grandjean Research Group – University College London, UK

Remote

PI: Louis Grandjean, M.D., Ph.D

- **Hospital Wastewater Metagenomics Project**: carried out taxonomic identification of bacterial species from long reads and MAGs; identified pathogen reads, assembled specific pathogen genomes and evaluated their quality; performed antibiotic resistance genes identification and statistical analysis from reads and assemblies to assess abundance consistency across different temporal and spatial sampling points. Extensively used Bash, Python and R.

Research Internship

Feb 2024 – Dec 2024

Laboratory of Bioinformatics and Molecular Biology – Peruvian University Cayetano Heredia, Peru

PI: Mirko Zimic, Ph.D

- **Soil Metagenomics Project**: executed basecalling of long reads and genome assembly; performed functional annotation of protein sequences; identified mobile genetic elements like

plasmids and phages and analyzed associated genes; conducted statistical analysis of enrichment and conservation of metabolic pathways.

- **Oral Microbiome Metagenomics Project:** conducted taxonomic identification, metagenome assembly from short reads, binning refinement and quality assessment; performed normalization, differential abundance analysis and clustering based on taxon abundance; statistically analyzed the abundance of resistance genes and their association with mobile genetic elements.
- **The Effect of Thermodynamic Pressure on Genome Evolution Project:** carried out DNA evolution simulation analyses through Python programming, considering the energy of neighboring base pairs according to the Boltzmann distribution.

Research Internship

Apr 2024 – Dec 2024

Laboratory of Molecular Evolution of Biota in Peru – National University of San Marcos, Peru

PI: Pedro Romero, Ph.D

- Studied evolution of terrestrialization from limited short-read sequencing data in three Amazonian species of the genus *Pomacea*: read & assembly QC, k-mer distribution analysis, gene annotation, variant calling and retrieval of specific genes (perivitellin duplicates and calcium-binding protein) associated with terrestrialization.

Research Internship

Aug 2023 – Jan 2024

Laboratory of Bioinformatics and Molecular Biology – Peruvian University Cayetano Heredia, Peru

PI: Patricia Sheen, Ph.D

- Performed PCR, DNA & plasmid extraction, cloning, ligation, digestion, transformation, electrophoresis and protein purification.

Research Internship

Feb 2023 – Jul 2023

Laboratory of Marine Biology – Peruvian University Cayetano Heredia, Peru

PI: Pedro Romero, Ph.D

- Executed phylogenetic analysis using Bayesian and ML methods on mitochondrial sequences from the gecko species *Phyllodactylus reissii*.

AWARDS & FUNDING

Academic Excellence Scholarship – two consecutive years (2024 & 2025) \$7 000/year
Peruvian University Cayetano Heredia

- Awarded twice in succession for ranking 1st in the annual merit order of the Faculty of Sciences and Engineering; covers 75% of tuition fees each year.

Research Grant 87234

\$120 000

**National Council for Science, Technology, and Technological
Innovation (CONCYTEC), Peru**

2024

- Conceptualized and authored the bioinformatics and molecular-biology section of the proposal “*The Effect of Thermodynamic Pressure on Genome Evolution*”. PI: Mirko Zimic, Ph.D

Student Mobility Support Fund
Institutional Relations and Internationalization Office
(DURIN), Peruvian University Cayetano Heredia

\$3 000
2024

- Selected through competitive evaluation by the University's International Relations Office to support an international research rotation under Resolution No. RESOR-SEGEN-UPCH-2024-CU-0201.

CERTIFICATIONS

Genomic Data Science Specialization (54 h) Jul 2024
Johns Hopkins University – Coursera

- Bioinformatics algorithms, command-line, Python, R and statistical methods for genomic and transcriptomic data

Navigating Genetic Variation: A Population Genomics Workshop (15 h) Apr 2024
University of San Martín de Porres – Barreiro Lab, University of Chicago

- R and Bash for genetic diversity analysis and functional genomics: GWAS, DEGs, and eQTL analysis.

Genomics & Metagenomics: Sequencing & Bioinformatics Analysis (30 h) Dec 2023
La Molina National Agrarian University

- Assembly, annotation and R-based analysis of genomic and metagenomic data.

RELEVANT COURSEWORK

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- **Bioinformatics I** (2024, 48h) — alignment algorithms, substitution matrices, phylogenetic trees, molecular markers, transcriptomics and molecular modeling. Project: **Computational Identification and Curation of Microsatellite Markers in *Ipomoea batatas***.
 - **Bioinformatics II** (2024, 48h) — comparative genomics, epigenomics, proteomics, metabolomics and multi-omics integration. Project: **Syntenic and Conservation of Metabolic Pathways in Photosynthetic Organisms**.
 - **Computing** – Introductory Python (2023, 32h).
 - **Advanced Programming** – Advanced Python (2025, 48h).
 - **Scientific Computing** (2024, 48h) — R/RStudio for data wrangling, visualization and machine learning.
 - **Computer Vision** (2025, 64h) — CNNs, autoencoders, GANs and vision transformers; implemented with PyTorch.
 - **Pattern Recognition** (2025, 48h) — classical ML (k-NN, SVM, decision trees) and introductory deep learning for image signal classification.
 - **Introduction to Machine Learning** (2025, 48h) — supervised/unsupervised learning, model evaluation, regularization and hands-on deep learning with TensorFlow.
 - **Parallel and Distributed Computing** (2025, 48h) — POSIX threads, OpenMP, MPI, Cloud Computing, Docker/Kubernetes and distributed ML training.
 - **Programming Language Paradigms** (2025, 48h) — C++, HTML/CSS/JavaScript and OOP concepts; compared imperative, functional and object-oriented paradigms.

CURRENTLY TAKING

- **Differential Equations** (2025, 80 h) — first and higher-order ODEs, modeling, Laplace transforms, numerical solutions, Fourier series & boundary-value problems.
- **Vector Calculus for Engineering** (2025, 80 h) — vector algebra, coordinate systems, multivariable differentiation (gradient, optimization, Lagrange multipliers) and integration, Green-Stokes-Gauss theorems.
- **Molecular Ecology** (2025, 64 h) — molecular evolution, phylogenetic reconstruction, phylogeography, genetic diversity & haplotype networks, divergence times, population genomics.

GITHUB REPOSITORIES

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- **virus2tree** (2025) — A new Read2Tree version for viral phylogenomics that reconstructs evolutionary history from assemblies and read sets: builds an OMA Standalone reference from NCBI assemblies, bulk-downloads SRA read sets, maps reads to that reference with optional deduplication (czid-dedup) and downsampling (rasusa), and gathers statistics from ortholog groups and read sets.
 - **Mosaicome** (2025) — Project born out of the Sedlazeck Lab Structural Variation Hackathon: results for structural-variant (SV) discovery and benchmarking (Sniffles2, cuteSV), truth-set curation and evaluation (Truvari), pangenome/graph alignment experiments (minimap2, vg), structural-variant QC, visualization, and reporting.
 - **synteny_viridiplantae_project** (2024) — R+Python scripts for macro and micro-synteny of conserved gene clusters across four plant genomes representing photosynthetic, parasitic and carnivorous species; detection of collinear synteny blocks with MCScanX, ortholog prediction with Orthofinder, functional annotation with EggNOG Mapper, pathway enrichment analysis with DAVID and PANTHER, and visualization with pygenomeviz, synvisio and ggplot2.
 - **wastewaters_project** (2024) — Figures and Rmd scripts to monitor pathogen abundance and antibiotic resistance genes in hospital wastewaters based on RGI screening and Kraken2/Sylph taxonomic profiling.
 - **oral_microbiome** (2024) — Figures and Nextflow+Rmd pipeline for shotgun oral-microbiome data: taxonomic profiling with Kraken2-Bracken, differential abundance with LEfSe, and beta-diversity via PCoA. Detects and quantifies antibiotic-resistance genes with RGI and evaluates resistome shifts using MaAsLin2 and ANCOM-BC2.
 - **soil_microbiome** (2024) — R+Python workflows for soil metagenomes (plasmid, viral, and chromosomal fractions): QC and depth summaries (mosdepth), functional annotation (EggNOG-mapper), diversity/ordination and differential analysis (vegan, MaAsLin2), and reproducible Rmd figures/tables for abundance patterns and beta-diversity.
 - **GenomicsFreedom** (2025) — Used HTML/CSS/JavaScript and Flask to create an interactive genomics workbench: GC and sequence stats, k-mer motif analytics, motif search, MSA and distance matrices settings and visualization, and MUSCLE-based phylogenetic trees.
 - **dnaStatsApp** (2025) — Interactive Streamlit dashboard for batch FASTA/FASTQ uploads, GC-content, codon-bias & k-mer profiling, motif search, MSA-based phylogenetic trees, sequence conservation analysis and Plotly visualizations.
 - **esm3** (2025) — Jupyter notebooks for fine-tuning the protein language model ESM-3 on antibody-antigen affinity prediction: zero-shot benchmarking, embedding extraction, supervised transfer learning with a lightweight regression head, and Streamlit deployment for interactive inference.

- **TextMineML** (2025) — NLP pipeline for Spanish biomedical literature: spaCy preprocessing, TF-IDF/word2vec/SBERT embeddings, K-means/HDBSCAN clustering, topic modeling, and UMAP/PCA cluster visualizations.
- **ScrapeFlow** (2025) — Selenium + OpenMP framework that scrapes multiple Peruvian e-commerce sites in parallel, retrieves product information and presents ordered data in an interactive and user-friendly UI.

POSTERS & PRESENTATIONS

- Ruiz-Ruiz, M.I., **Chalco, A.**, Ramírez, R., & Romero, P.E. (2025). *Molecular insights into the role of perivitellins in the evolution of terrestrial oviposition in Pomacea Perry (1811) (Gastropoda: Ampullariidae)*. Virtual poster presented at the **Wellcome Connecting Science – Understanding Life: Using Large-scale Biodiversity Reference Genomes**, Hinxton, UK. *Repository: Poster (GitHub)*
- Paulin, L., Baker, D., Woycicki, R., Mondal, R., Fabián, G., **Chalco, A.**, Mahmoud, A., Baig, N., Subramaniam, G., Saleem, A., Monahan, G., Krol, J., Yang, X., Ghohabi Esfahani, N., & Fu, Q. (2025). Remote team presentation of the Mosaicome Project at the **7th Structural Variation Hackathon**, Baylor College of Medicine, US, Aug 27–29, 2025. *Repository: Mosaicome (GitHub)*.
- Ruiz-Ruiz, M.I., **Chalco, A.**, Ramírez, R., & Romero, P.E. (2025). *Molecular insights into the role of perivitellins in the evolution of terrestrial oviposition in Pomacea Perry (1811) (Gastropoda: Ampullariidae)*. Poster presented at the **III National Congress on Life Sciences - XXX Scientific Meeting of ICBAR**, Universidad Nacional Mayor de San Marcos, Lima, Peru. *Repository: Poster (GitHub)*
- **Chalco, A.**, Hurtado, B., & Orellana, J. (2025). *Transfer learning on the ESM3 protein generative model to predict antibody–antigen binding affinity*. Poster presented at the **Biomedical Engineering Expo 2025-I**, Faculty of Sciences and Engineering, Peruvian University Cayetano Heredia (UPCH), Lima, Peru. *Repository: Poster (GitHub)*

MANUSCRIPTS

TO BE SUBMITTED IN 2026

- Agostinho D., **Chalco A.**, Majidian S., Zheng X., & Sedlazeck F. *Bypassing assembly: Rapid phylogenomic analysis for viral surveillance and metagenomic profiling*. **Contribution:** I created the virus2tree software, and designed and performed computational analyses.
- Zimic A., Balta M., **Chalco A.**, Taquiri-Díaz D., Soto C., Romero O., Orellana J., León C., De-Stefano L., & Zimic M. *Resilience and functional reconfiguration of the Persea americana rhizosphere microbiome following anaerobic digestate application* **Contribution:** I designed and performed computational analyses, and contributed to manuscript writing.
- Ruiz-Ruiz M.I., **Chalco A.**, Ramírez, R., & Romero P. *Molecular insights into the role of perivitellins in the evolution of terrestrial oviposition in Pomacea Perry (1811) (Gastropoda: Ampullariidae)*. **Contribution:** I designed and performed computational analyses, and contributed to manuscript writing.
- Lopez-Ramos R., **Chalco A.**, Taquiri-Díaz D., Romero O., Zimic M., & Pajuelo M. *Oral mi-*

crobiome profile in early childhood caries in Peruvian preschoolers. Contribution: I designed and performed computational analyses, and contributed to manuscript writing.

- Omaston A., Mascaro L., Merino L., Lawson G., Tan G., Romero O., Taquiri-Díaz D., Avelaneda P., Apaza C., Orellana J., **Chalco A.**, Leon Palomino C., Sheen P., Zimic M., Pajuelo M., & Grandjean L. *Multicenter Hospital and Environmental Wastewater Sampling for Antimicrobial Resistance and Novel Pathogen Surveillance. Contribution:* I designed and performed computational analyses.
- Hurtado B., **Chalco A.**, Orellana J., Alvarado J., Meza M., & Zimic M. *Transfer learning on the generative protein model ESM-3 to predict antibody-antigen affinity. Contribution:* I designed and performed computational analyses, and contributed to manuscript writing.
- Collazos V., Aricoche K., Atachahua K., Apaza C., **Chalco A.**, Hurtado B., Sheen P., & Zimic M. *Thermodynamic Constraints on GC-Content Evolution during Serial Polymerase-Mediated Amplification of Bacterial DNA. Contribution:* I designed and performed computational analyses.

EXTRACURRICULAR ACTIVITIES

Directive Board Member

Feb 2024 – Dec 2024

Journal Club – Peruvian University Cayetano Heredia

- Recruited presenters for weekly scientific discussion clubs on topics related to molecular biology, coordinated leadership talks for students and started the establishment of journal clubs in schools to stimulate interest in science.

Coordinator

Feb 2023 – Jul 2023

Ecology, Evolution & Conservation Group – Peruvian University Cayetano Heredia

- Organized lectures and workshops on molecular ecology.

Leader

Feb 2015 – Nov 2022

Pathfinder Club (International youth program)

- Designed educational activities, camps and community projects for youth; strengthened leadership and communication skills.

SKILLS

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- **Languages:** Spanish (native speaker), English
 - **Programming:** Bash, R/RStudio, Python, C++, HTML, CSS, JavaScript
 - **Toolbox:** VS Code, Cursor AI, Nextflow, SSH, Regular Expressions, Docker, Kubernetes, LaTeX, Git/GitHub

REFERENCES

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- Ph.D. **Fritz Sedlazeck*** – Professor at the Human Genome Sequencing Center, Baylor College of Medicine, Houston, USA

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- Ph.D. **Mirko Zimic*** – Professor of Bioinformatics and Molecular Biology, Peruvian University Cayetano Heredia, Peru
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- Ph.D. **Felipe Yon** – Professor of Molecular Ecology, Peruvian University Cayetano Heredia, Peru
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- M.D., Ph.D. **Louis Grandjean** – Professor of Infectious Diseases, University College London, London, UK
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- Ph.D. **Pedro Romero** – Professor of Molecular Ecology, National University of San Marcos, Peru
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- Ph.D. **Sina Majidian** – Postdoctoral Fellow, The Johns Hopkins University, Baltimore, MD, USA
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* *These are my two selected referees.*